NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 11/01 /2026
Time: 3 Hrs | GTM – 15 (TBT-2) | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
C	C	C	B	A	B	C	A	C	C
11	12	13	14	15	16	17	18	19	20
B	B	A	D	C	D	C	B	A	C
21	22	23	24	25					
25	14	123	14	124					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	C	B	D	D	D	A	D	D	D
36	37	38	39	40	41	42	43	44	45
C	C	A	C	B	B	B	C	A	D
46	47	48	49	50					
1	4	4	7	3					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	D	C	B	D	A	A	C	C	B
61	62	63	64	65	66	67	68	69	70
C	A	D	D	B	B	C	D	B	C
71	72	73	74	75					
11	250	1	36	20					

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APEX
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HINTS
PHYSICS

1. The given expression is $P = EL^2m^{-5}G^{-2}$

Dimensions of energy, $E = \left[ML^2T^{-2}\right]$

Angular momentum, $L = \left[ML^2T^{-1}\right]$

Mass, $m = \left[M\right]$

Gravitational constant, $G = \left[M^{-1}L^3T^{-2}\right]$

Substituting dimensions of each term in the given expression,

$$P = \left[ML^2T^{-2}\right] \times \left[ML^2T^{-1}\right]^2 \times \left[M\right]^{-5} \times \left[M^{-1}L^3T^{-2}\right]^{-2}$$

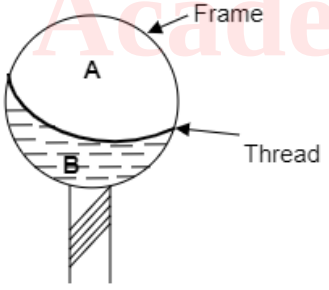
$$= \left[M^{1+2-5+2}L^{2+4-6}T^{-2-2+4}\right] = \left[M^0L^0T^0\right]$$

2. $E = \frac{U}{2} = -K = -\frac{GMm}{2r}$

3. For negative acceleration, the $x-t$ graph moves downward. But the car is moving in positive direction as the position coordinate is increasing in the positive direction.
Thus, the statement given in option (C) is correct, rest are incorrect.

4. Since, metal is denser than ice, water displaced will be more than the water formed on melting, so the level of water falls.

6. When film A is punctured, film B will try to minimise its area. So, thread will become concave towards A.



8. Thermal resistance, $R = \frac{l}{KA}$

$\therefore \quad R \propto l$

$$i = \frac{T_1 - T_2}{R} \Rightarrow i \propto \frac{1}{l}$$

$\therefore \quad \frac{i_1}{i_2} = \frac{l_2}{l_1} = \frac{2r}{\pi r} = \frac{2}{\pi}$

9. $T = \frac{pV}{nR}$,

$T' = \frac{(2p)(2V)}{nR} = 4T$

$v_{avg} \propto \sqrt{T}$

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So, if temperature becomes 4 times, average speed will get doubled.

11. First velocity of the iron sphere increases and after sometime becomes constant. Hence, accordingly first KE increases and then becomes constnat which is best represented by (B)

12. Use formula MI

13. As the reflected pulse is not inverted, the second medium is rarer. $v_2 > v_1 \Rightarrow \lambda_2 > \lambda_1$ [$\because$ Frequency doesn't change]

14. A vector is independent of choice of reference frame. If a vector is shifted parallel to itself, it does not change. If a vector is rotated by an angle other than a multiple of 2π, it changes.

15. NCERT exercise problem

16. $I_x = I_y = \dfrac{MR^2}{4}$

If radius of gyration is K_1, then,

$$I_x = I_y = MK_1^2 \Rightarrow K_1 = \frac{R}{2}$$

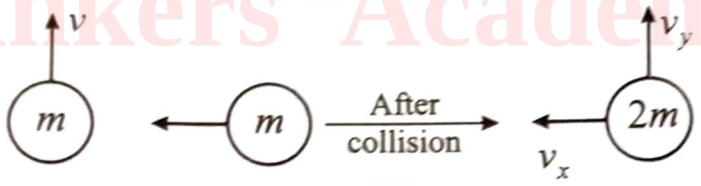
Similarly,

$$I_z = \frac{MR^2}{2} = MR_{K_2}^2 \Rightarrow K_2 = \frac{R}{\sqrt{2}}$$

Hence, $K_1 \neq K_2$

17. Mean free path is independent of temperature and relaxation time decreases at temperature increases.

18.



$$2mv_x = mv$$

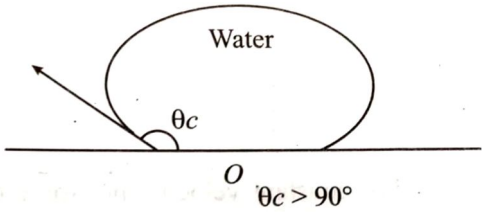
$$\Rightarrow v_x = \frac{v}{2}$$

$$v_y = \frac{v}{2}$$

$$v_{net} = \sqrt{v_{x2} + v_{y2}} = \frac{v}{\sqrt{2}}$$

So, path will be elliptical

19.



For water oil interface

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20. $n_1 = \frac{v}{2l_2}, n_2 = \frac{v}{2l_2}, n_3 = \frac{v}{2l_3}$

For complete wire, $n = \frac{v}{2l}$

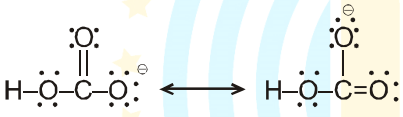
$l = l_1 + l_2 + l_3 + ...$

$\Rightarrow \frac{v}{2n} = \frac{v}{2n_1} + \frac{v}{2n_2} + \frac{v}{2n_3} + ...$

$\Rightarrow \frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} + ...$

CHEMISTRY

29. N_3^-, $(CNO)^-$ and $(NCN)^{2-}$ all have same number of electrons i.e., 22; so all are isoelectronic with CO_2 which also has 22 electrons.

30. 

$\therefore x > y = z$

31. Higher the metallic character, greater will be the basic character of its oxide as ΔE_n (difference in electronegativities between element and oxygen) increases.
So, $NiO < MgO < SrO < K_2O < Cs_2O$

40. Differential extraction is based on the principle of solubility in two different solvents.

41. Kjeldahi method is not applicable for compounds containing nitrogen in nitro, and azo groups and nitrogen in ring, as N of these compounds does not change to ammonium sulphate under these conditions. Hence only aniline can be used for estimation of nitrogen by Kjeldahi's method.

42. Boiling point depends on intermolecular attractive forces. Among isomeric alkanes, linear alkanes have more boiling point that branched alkanes.

43. Reactivity towards Ar-Se $\propto +M, +H + I \propto \frac{1}{-M, -H, -1}$

$CH_3(+H), -CF_3(-H), -OCH_3(+M), -N(Me)_2(+M)$

$e > d > a > b > c$

44. In isomers of hydrocarbons heat of combusition depends upon their stabilities.
As the stability increases heat of combustion decreases stability order.

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